

DEBOJYOTI DEY

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SUMMARY

Product-driven Full Stack Engineer with **3+ years of experience** specializing in real-time AI platforms, multi-agent LLM orchestration, and low-latency voice applications. Experienced in building asynchronous FastAPI architectures, fine-tuning classification models, and developing VoIP/telephony systems (LiveKit, Twilio). Bridging the gap between robust engineering, Kubernetes infrastructure, and product strategy to deploy high-scale AI products.

EXPERIENCE

Full Stack Engineer (AI / Real-time Systems)

Jan 2026 – Present

Omnis AI — AI-native Operating System for the Legal Industry

Remote

- Architected a multi-agent orchestration engine using **FastAPI** and **Google Agent Development Kit (ADK)** to dynamically provision and version customizable AI voice and chat agents.
- Engineered a scalable VoIP & telephony pipeline by integrating **Twilio (PSTN)** with **LiveKit (WebRTC)** and **Ultravox**, enabling real-time, low-latency (<500ms) voice calls with AI agents.
- Integrated **LiteLLM** to route queries across multiple LLM providers (OpenAI, Gemini, Grok, DeepSeek) with automatic fallback and retry logic, reducing API failure rates by **40%**.
- Implemented dynamic tool discovery using **Model Context Protocol (MCP)** and PostgreSQL/Redis registries, and built a guardrails system using **pgvector** for intent-steering.
- Developed high-concurrency async endpoints utilizing **SQLAlchemy 2.0 (Async)** and **Celery** to handle WebSockets, IVR media stream handoffs, and background call transcription workflows.

Co-Founder & Tech Lead (Backend / AI / Product)

Aug 2023 – Dec 2025

AtoZDebug — AI Automation & Software Development Agency

Remote / India

- Led end-to-end development of AI-driven SaaS products, translating business goals into robust backend architectures using **Node.js**, **Python**, **TypeScript**, **PostgreSQL**, and **MongoDB**.
- Built LLM-powered multi-agent task orchestration pipelines using **LangChain**, **CrewAI**, and **LangGraph**, reducing manual processing times by **60%** for international clients.
- Designed custom semantic search and Retrieval-Augmented Generation (RAG) patterns to address context-window limits, improving response accuracy.
- Managed a cross-functional team of 5 engineers using Agile/Scrum methodologies, accelerating time-to-market and increasing client retention across 3 core projects.
- Translated user analytics and feedback into iterative product features, ensuring high-quality product delivery.

PROJECTS

BluBuddy.io — Real-time Conversational AI Platform

Built a real time AI conversation platform — Python WebSocket backend, React on the frontend, with LangGraph and LangChain handling the multi agent orchestration underneath. Conversation retrieval stays fast even at volume thanks to MongoDB Vector Search. Billing runs through Paddle for global subscriptions, all hosted on AWS. So far it's handled 50,000+ live interactions and bumped conversion 35%.

SmartInbox.ai — AI Email Agent & Prioritization Engine

Built an AI email tool that routes and prioritizes your inbox — Python backend, Next.js on top. Pulls live data through the Gmail API, then LlamaIndex and Pinecone handle the semantic search to flag what's actually urgent. Stripe runs the billing. It's cut daily email time by 40% for 2,000+ active users.

AskMate AI — *Local RAG Study Assistant*

Developed a cross-platform mobile app using React Native and FastAPI. Built custom local document ingestion, processing, and vector indexing (ChromaDB) to provide context-aware answering and interactive flashcard generation.

SKILLS

AI & LLM Orchestration: Google Agent Development Kit (ADK), LiteLLM, Model Context Protocol (MCP), LangGraph, LangChain, CrewAI, Multi-Agent Systems, RAG, Prompt Engineering.

Real-Time & Telephony: LiveKit & LiveKit Agents, WebRTC, Ultravox SDK, Twilio SDK (PSTN, SIP Trunking, TwiML), WebSockets.

Full-Stack Engineering: Python (FastAPI/Flask), Celery, Uvicorn, Node.js, TypeScript, Next.js, React, React Native, REST APIs, SQLAlchemy 2.0 (Async), Alembic.

Databases & DevOps: PostgreSQL (pgvector), Redis, MongoDB, ChromaDB, Docker, Kubernetes (EKS), ArgoCD, Amazon ECR, Sealed Secrets, AWS, Git.

Product & Strategy: MVP Development, PRDs, Roadmap Prioritization, Agile/Scrum, User Research, KPI Tracking.

EDUCATION

B.Tech in Computer Science and Engineering	2020 – 2023
Barak Valley Engineering College, India	

Diploma in Computer Engineering	2017 – 2020
Nowgong Polytechnic, India	

CERTIFICATIONS

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| • Product Manager Fellowship , NextLeap | Dec 2025 – Mar 2026 |
| • FastAPI Certification , Udemy | 2025 |
| • Advanced Certification , Make | 2025 |